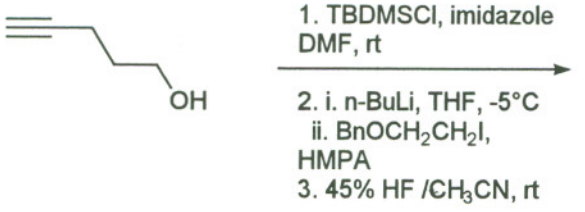
Organic Cume
April 5, 2002

Source:
"A Formal Total Synthesis of (±)-Cephalotaxine Using Sequential N-Acyliminium Ion Reactions" Yuji Koseki, Hiroto Sato, Yumi Watanabe, and Tatsuo Nagasaka, *Org. Lett.*, **2002**, *4*, 885-888.

<u>You will need to answer any **three** of the four questions</u>. You **do not** need to show stereochemistry. Be sure to show step-by-step mechanisms for the indicated reactions, i.e. don't skip any steps.

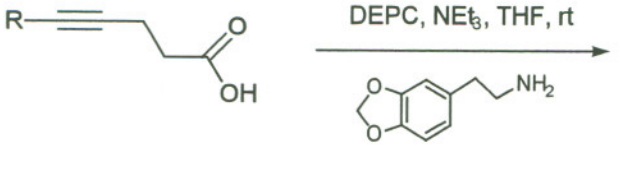
1.

a.



1

b.



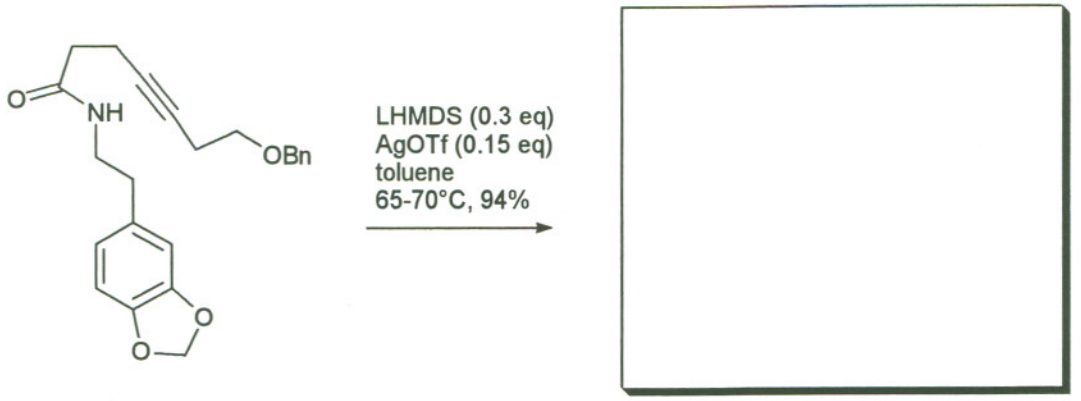
2

c. What is the role of DEPC?

d. What is the structure of DEPC?

2.

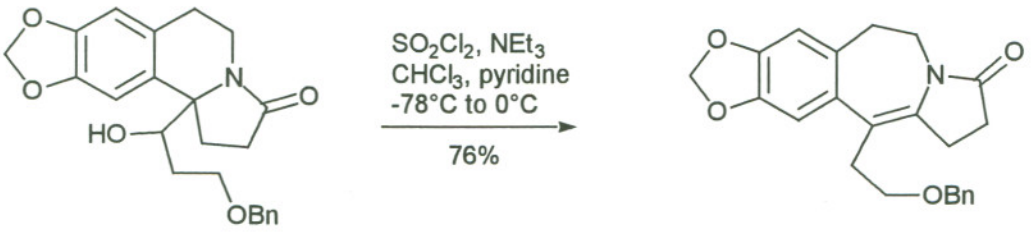
a.



3

b. Propose a step-by-step mechanism for the above reaction.

3.



Propose a step-by-step mechanism for the above reaction.

4.

a.

4

5 (pentacyclic product)

b. What is the role of Dess-Martin periodinane?

c. What is the structure of Dess-Martin periodinane?

d. Propose a step-by-step mechanism for the reaction **4** -> **5**.